

Muhammad Ibrahim

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SUMMARY

Electrical and Computer Engineering M.Sc. candidate with hands-on experience in wireless communication research, L1/baseband-oriented simulation, optimization algorithms, C/C++, Python, MATLAB, Linux, Git, and embedded systems. Strong foundation in wireless communication, MIMO, beamforming, channel modeling, OFDMA-based systems, receiver-side performance analysis and real-time embedded workflows.

EXPERIENCE

WiCoNS | RA

Sep. 2024 – Present

NSERC-Funded Research Projects

Winnipeg, Canada

- **Project:** Wireless System Optimization and Validation
Modeled multi-user wireless systems and evaluated optimization algorithms for beamforming, phase-shift design, achievable-rate maximization, convergence behavior, numerical validation, parameter sensitivity, and baseline comparison.
- **Project:** Deep RL for Networked Wireless Systems
Proposed a hierarchical deep reinforcement learning approach for communication-graph selection in multi-agent consensus systems, analyzing convergence stability, communication overhead, robustness, and performance tradeoffs.

ESDAC | Embedded ML Intern

May 2022 – Aug. 2022

Embedded AI and Edge Deployment

Islamabad, Pakistan

- **Project:** Jetson Nano Edge AI
Deployed real-time object detection and classification models on NVIDIA Jetson Nano under Linux-based hardware constraints. Optimized models with TensorRT, improving inference efficiency by 25–40% while analyzing latency, memory limits, and deployment tradeoffs.

Burraq | AI and Electrical Engineer

Mar. 2022 – Dec. 2023

Student-Led UAV Engineering Team

Islamabad, Pakistan

- **Project:** Autonomous Pesticide Spraying UAV
Worked on electrical and embedded-system integration for a targeted pesticide-spraying UAV, including sensor interfaces, camera integration, power electronics, telemetry review, and field-test debugging.
- **Project:** Drowning Detection and Life-Jacket Delivery UAV
Worked on UAV system integration for a vision-assisted drowning detection and life-jacket delivery prototype using cameras, embedded controllers, telemetry logs, sensor readings, UART, I2C, GPIO, and ADC.

PROJECTS

Dynamic Hybrid Vision Transformer

Python, Deep Learning, Computer Vision, Transformers

- Implemented a hybrid vision transformer architecture for image classification by combining transformer-based representation learning with convolutional inductive biases and dynamic feature aggregation for efficient model evaluation.

TCP Traffic and Session Analysis Using Wireshark

Wireshark, TCP/IP, Networking

- Captured and analyzed TCP/IP client-server traffic, verified protocol behavior, identified endpoints and TCP sessions, and quantified traffic using packet rate, throughput, and session-level analysis.

RISC-V Processor Design and FPGA Implementation

Verilog, FPGA, RISC-V, Digital Design

- Implemented a modular RISC-V processor datapath in Verilog with ALU, register file, program counter, instruction decoder, instruction memory, data memory, comparator, and top-level core modules.

ESP32 Telemetry and Device Monitoring Platform

ESP32, FreeRTOS, FastAPI, SQLite

- Built ESP32-S3 firmware with Wi-Fi reconnection logic, FreeRTOS tasks, NVS configuration storage, device-health metrics, command handling, telemetry reporting, REST endpoints, WebSocket updates, and integration tests.

EDUCATION

University of Manitoba

2024 – 2026 Expected

Master of Electrical and Computer Engineering, CGPA: 4.38/4.5

Winnipeg, Canada

National University of Sciences and Technology

2020 – 2024

Bachelor of Electrical Engineering, CGPA: 3.66/4.0

Islamabad, Pakistan

TECHNICAL SKILLS

Programming and Systems: C, C++, Python, MATLAB, Bash, Linux, Git, Docker, FastAPI, SQLite, PostgreSQL, Data Analysis

Wireless system: 5G/6G, MIMO, Beamforming, OFDMA, Channel Modeling, Optimization

Embedded and Tools: FreeRTOS, ESP32-S3, Raspberry Pi, Jetson Nano, TensorRT, Verilog, FPGA, Wireshark, UART, I2C, SPI, GPIO, ADC